CELLULAR GENETICS

# Module map

- Module 08: Cellular Genetics
- Connected lab: lab-09_cellular-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 08
Cellular Genetics

1. Chromosomes and cell cycle
2. Mitosis for growth and repair
3. Meiosis for gametes
4. Variation from recombination and assortment
5. Errors and genetic consequences

**Lab evidence**
lab-09_cellular-genetics.md

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# Learning objectives

- Relate chromosomes to DNA organization.
- Compare the purpose and products of mitosis and meiosis.
- Explain how meiosis reduces chromosome number.
- Describe how crossing over and independent assortment increase variation.
- Interpret chromosome number errors at an introductory level.

**OBJECTIVE 1**
Relate chromosomes to DNA organization.

**OBJECTIVE 2**
Compare the purpose and products of mitosis and meiosis.

**OBJECTIVE 3**
Explain how meiosis reduces chromosome number.

**OBJECTIVE 4**
Describe how crossing over and independent assortment increase variation.

**OBJECTIVE 5**
Interpret chromosome number errors at an introductory level.

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# Topic sequence

- Chromosomes and cell cycle: Chromosomes organize DNA so cells can divide accurately.
- Mitosis for growth and repair: Mitosis produces genetically identical daughter cells for growth and repair.
- Meiosis for gametes: Meiosis produces haploid gametes for sexual reproduction.
- Variation from recombination and assortment: Crossing over and independent assortment create genetic variation.
- Errors and genetic consequences: Nondisjunction and other errors can change chromosome number.

01

**Chromosomes and cell cycle**

Chromosomes organize DNA so cells can divide accurately.

02

**Mitosis for growth and repair**

Mitosis produces genetically identical daughter cells for growth and repair.

03

**Meiosis for gametes**

Meiosis produces haploid gametes for sexual reproduction.

04

**Variation from recombination and assortment**

Crossing over and independent assortment create genetic variation.

05

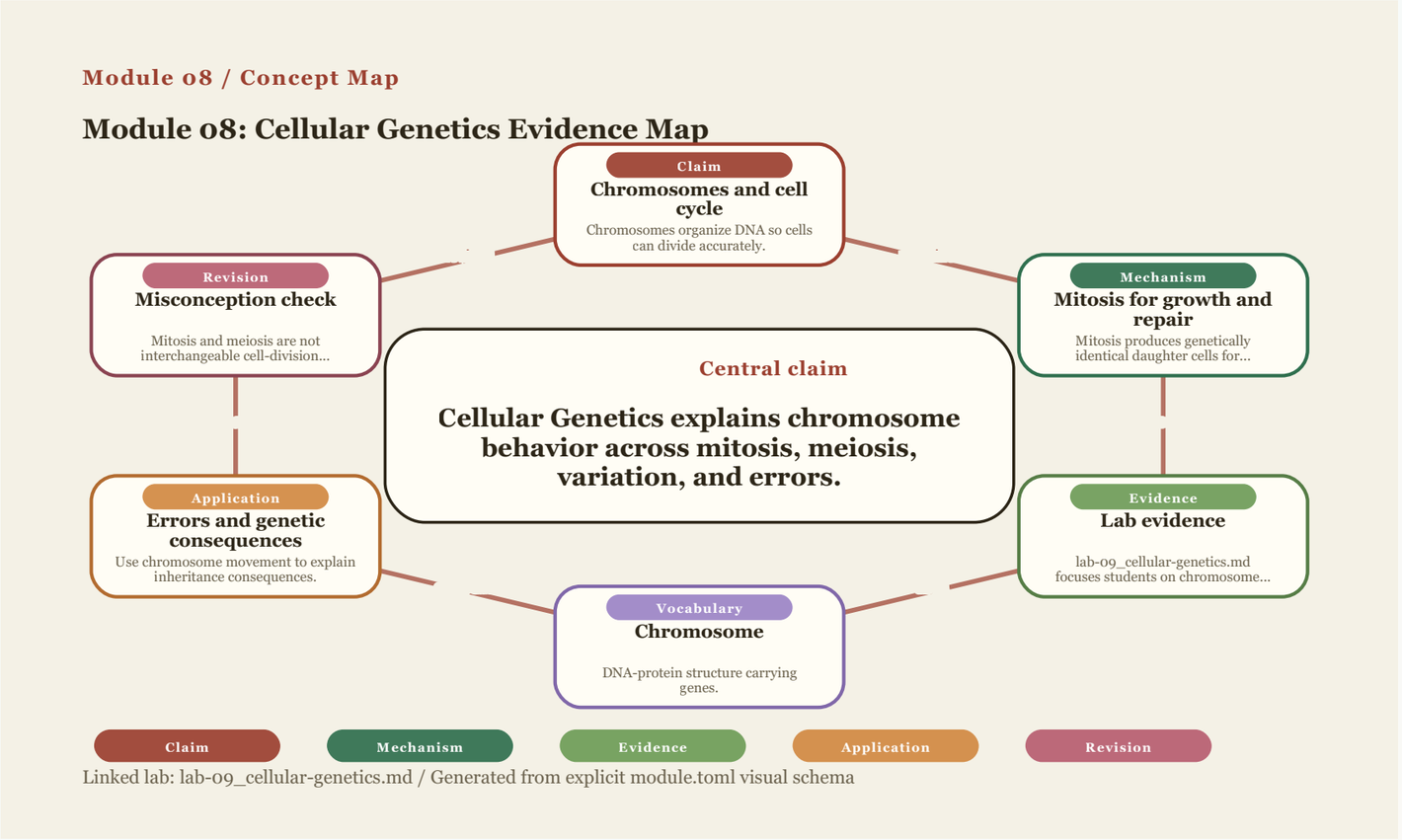
**Errors and genetic consequences**

Nondisjunction and other errors can change chromosome number.

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# Module 08: Cellular Genetics Evidence Map

- Module focus: Cellular Genetics
- Central claim: Cellular Genetics explains chromosome behavior across mitosis, meiosis, variation, and errors.
- Key nodes to connect: Chromosomes and cell cycle, Mitosis for growth and repair, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-09_cellular-genetics.md supplies an evidence surface for the map.

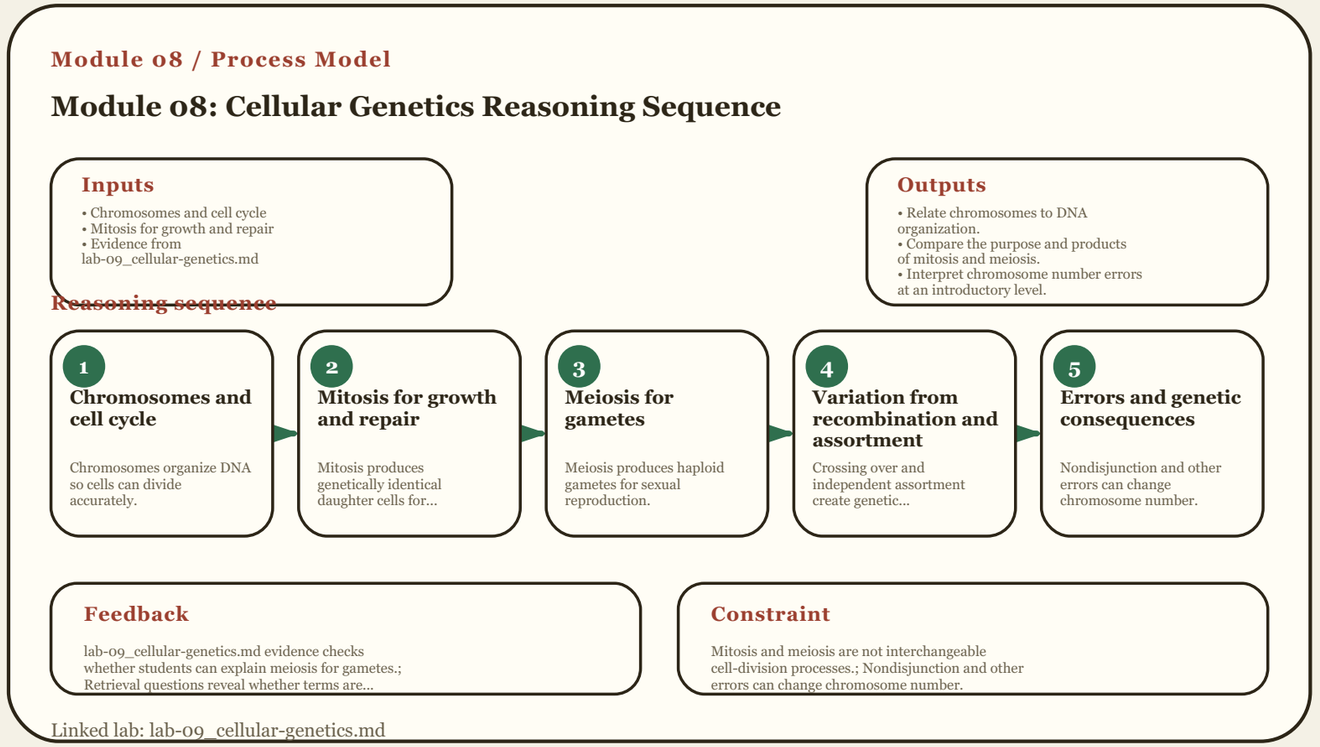



Linked lab: lab-09_cellular-genetics.md / Generated from explicit module.toml visual schema

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# Module 08: Cellular Genetics Reasoning Sequence

- Module focus: Cellular Genetics
- Trace the model: Chromosomes and cell cycle -> Mitosis for growth and repair -> Meiosis for gametes -> Variation from recombination and assortment
- Inputs become outputs: Chromosomes and cell cycle, Mitosis for growth and repair -> Relate chromosomes to DNA organization., Compare the purpose and products of mitosis and meiosis.
- Feedback check: lab-09_cellular-genetics.md evidence checks whether students can explain meiosis for gametes.
- Lab connection: lab-09_cellular-genetics.md gives students a process to observe or test.



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# Terms as evidence handles

- Chromosome: DNA-protein structure carrying genes.
- Cell cycle: Ordered sequence of cell growth, DNA replication, and division.
- Mitosis: Nuclear division producing identical daughter nuclei.
- Meiosis: Division process producing haploid gametes.
- Gamete: Haploid reproductive cell.
- Crossing over: Exchange of DNA between homologous chromosomes.

**CHROMOSOME**
DNA-protein structure carrying genes.

**CELL CYCLE**
Ordered sequence of cell growth, DNA replication, and division.

**MITOSIS**
Nuclear division producing identical daughter nuclei.

**MEIOSIS**
Division process producing haploid gametes.

**GAMETE**
Haploid reproductive cell.

**CROSSING OVER**
Exchange of DNA between homologous chromosomes.

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# Lab connection

- Lab file: lab-09_cellular-genetics.md
- Lab evidence should test or illustrate: Meiosis for gametes
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**
Chromosomes and cell cycle

**EVIDENCE**
lab-09_cellular-genetics.md

**CLAIM**
Relate chromosomes to DNA organization.

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# Contrast check

- Do not stop at: Chromosomes organize DNA so cells can divide accurately.
- Stronger explanation: Nondisjunction and other errors can change chromosome number.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Chromosomes organize DNA so cells can divide accurately.

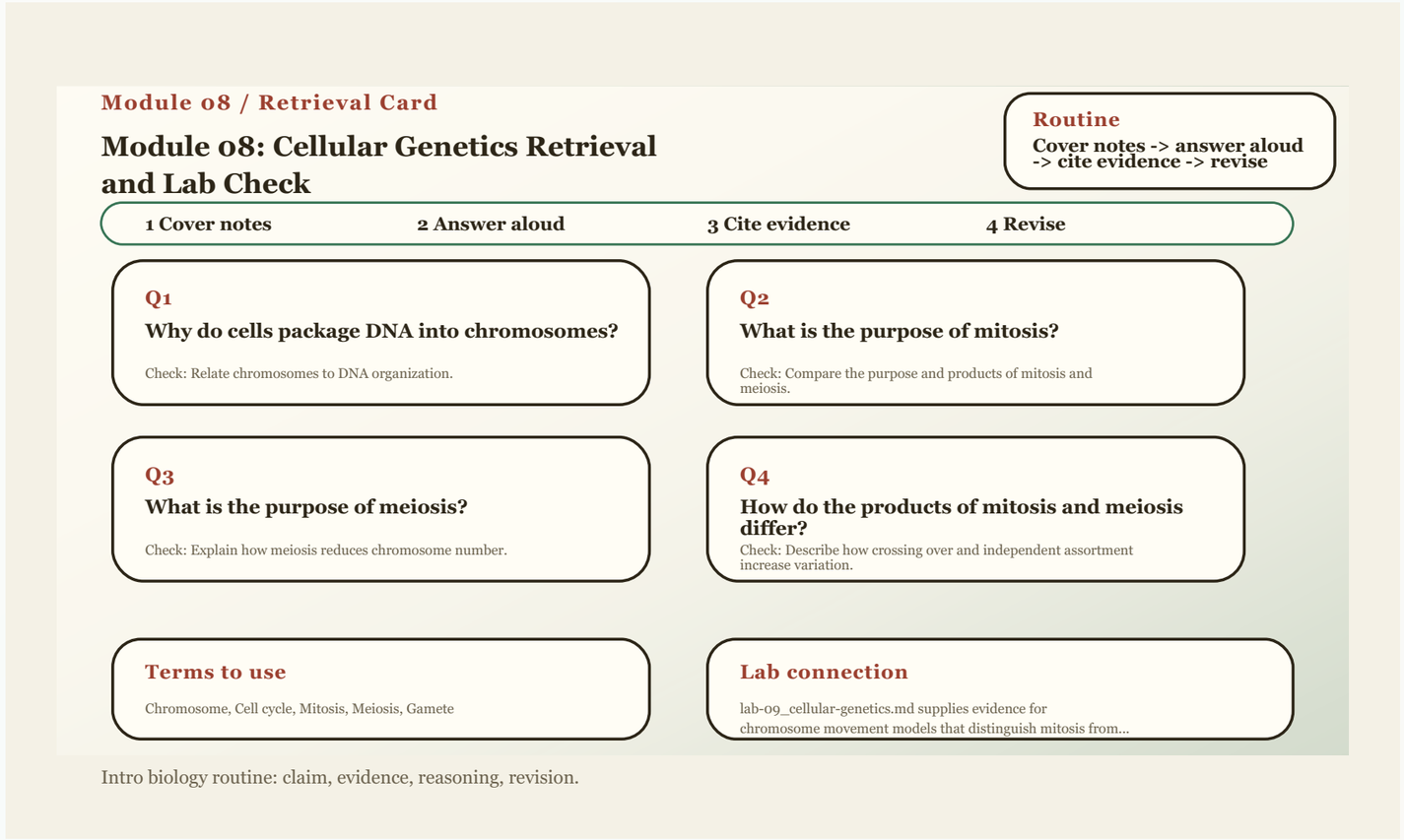
**DEEPER**

Nondisjunction and other errors can change chromosome number.

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# Module 08: Cellular Genetics Retrieval and Lab Check

- Module focus: Cellular Genetics
- Retrieval prompt: Why do cells package DNA into chromosomes?
- Answer check: Relate chromosomes to DNA organization.
- Required terms: Chromosome, Cell cycle, Mitosis
- Lab connection: lab-09_cellular-genetics.md; lab-09_cellular-genetics.md supplies evidence for chromosome movement models that distinguish mitosis from meiosis.




Intro biology routine: claim, evidence, reasoning, revision.

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# Practice quiz bridge

- 1. What is the main product of mitosis?
- 2. Meiosis reduces the chromosome number by using:
- 3. Crossing over and independent assortment both act to:
- 4. If chromosomes fail to separate correctly during meiosis (nondisjunction), a resulting gamete may have:

**QUESTION 1**

## Answer first

What is the main product of mitosis?

**QUESTION 2**

## Answer first

Meiosis reduces the chromosome number by using:

**QUESTION 3**

## Answer first

Crossing over and independent assortment both act to:

**QUESTION 4**

## Answer first

If chromosomes fail to separate correctly during meiosis (nondisjunction), a resulting gamete may have:

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# Synthesis and exit ticket

- Exit claim: explain chromosomes and cell cycle using evidence from lab-09_cellular-genetics.md.
- Must include: Chromosome, Cell cycle, Mitosis.
- Revision prompt: what evidence would change your explanation?

**CLAIM**
Chromosomes and cell cycle

**EVIDENCE**
lab-09_cellular-genetics.md

**REVISION**
What would change your mind?